

High Performance Vision-Guided Rocket Tracker

Motivation

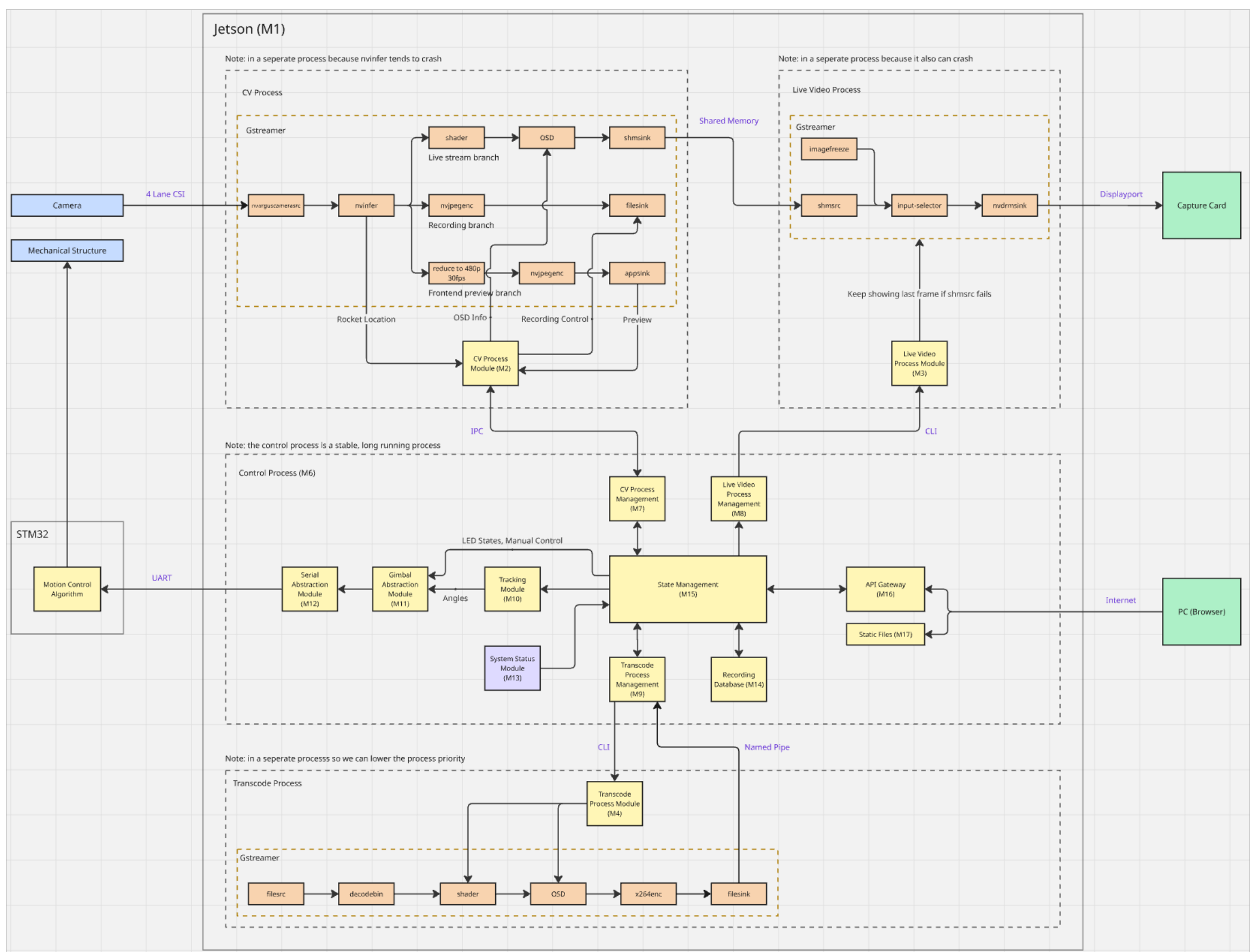
The Problem: Model rockets accelerate to extreme speeds within fractions of a second, making manual camera tracking virtually impossible. Existing commercial tracking systems are cost-prohibitive and lack the precision required for small, fast-moving targets. Consequently, rocketry teams lose critical visual data for mid-flight events like staging and parachute tangling.

The Solution: RoCam bridges this gap with an **autonomous, vision-guided tracking system**. By locking onto the rocket in real time, it actively commands a two-axis gimbal to deliver smooth **1080p video at 60 fps**.



RoCam Launch and Tracking Operations

System Architecture



Real-Time Vision and Control Pipeline

Edge Compute & Hardware. The implemented architecture follows a distributed multi-process edge-compute design on an NVIDIA Jetson. To ensure high reliability, CPU/GPU-heavy perception workloads are isolated from control and operator services via Inter-Process Communication (IPC). This strict fault isolation preserves deterministic command flow and API stability even if the vision pipeline encounters transient errors during launch-dynamic conditions.

Vision Pipeline. Image acquisition utilizes a GStreamer capture graph configured for 1920×1080 at 60 fps, feeding NVIDIA DeepStream for GPU-accelerated preprocessing. Detection relies on a YOLO network optimized with TensorRT, yielding low-latency normalized bounding boxes. A decoupled Live Video process parallelizes OSD telemetry overlay and HDMI output to guarantee continuous situational awareness without blocking inference.

Kinematic Control. The tracking module calculates image-plane target error (targeting frame center $cx = 0.5, cy = 0.5$) to generate proportional 2-axis gimbal actuation commands. These commands pass through a validated, state-gated pathway—enforcing rate limits and safety bounds—before transmission to an STM32 microcontroller via UART, ensuring deterministic, closed-loop responsiveness throughout boost and coast regimes.

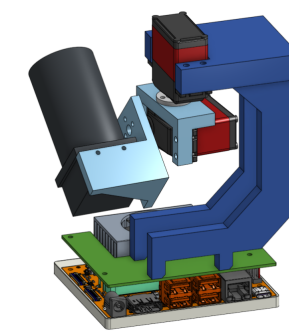
Backend & State Management. System orchestration is governed by a centralized State Machine (Disarmed, Armed, Tracking, Recording) that prevents invalid mode combinations and enforces hardware safety. A Flask-based API Gateway exposes a REST interface for offline-first LAN operation, managing mode transitions, asynchronous recording/transcoding workflows, and streaming SSE telemetry for live state and control-stream observability.

Software & Hardware Features



Web-Based Operator Interface

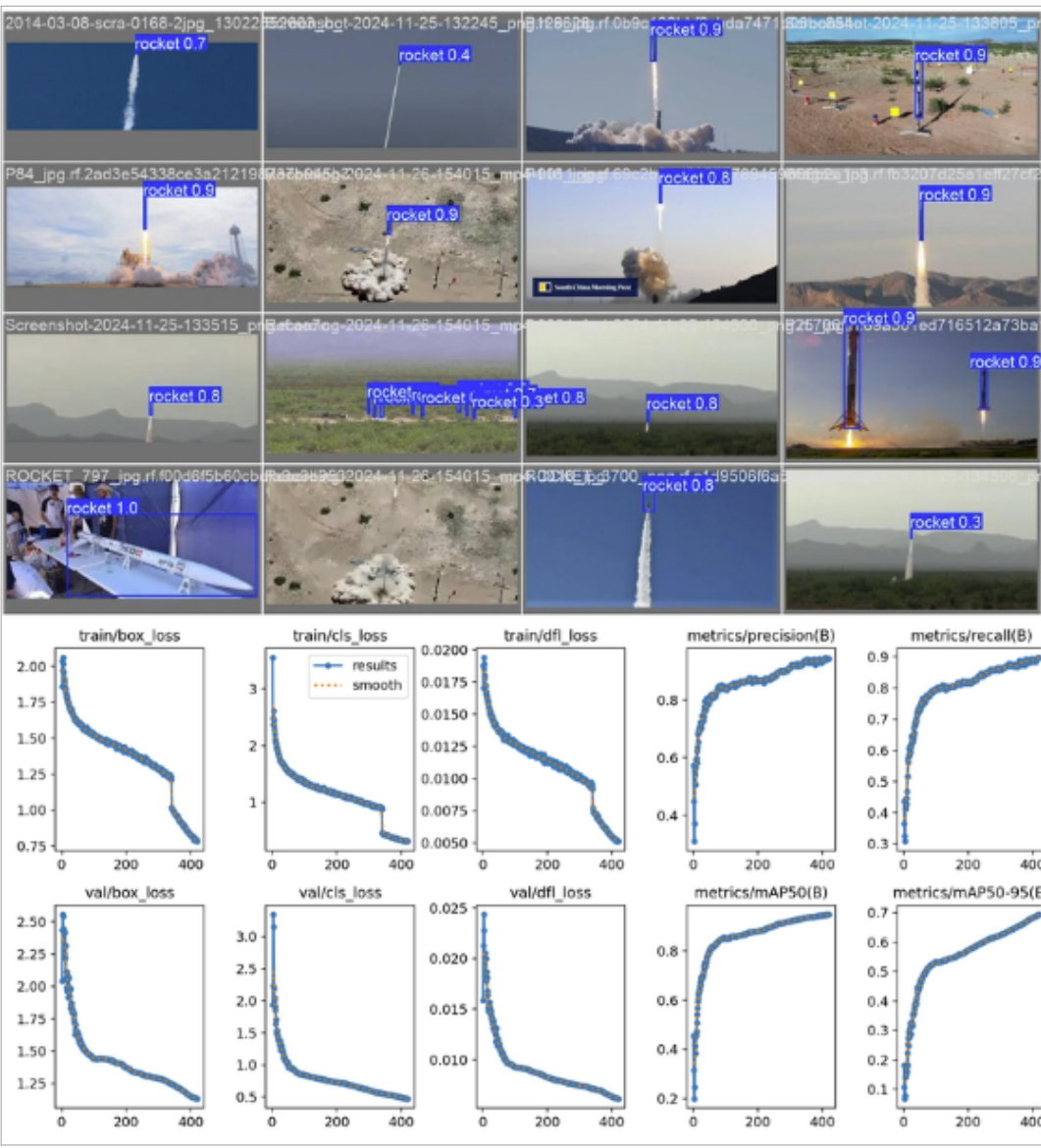
- **Real-Time Gimbal Control:** Low-latency control path between the operator interface and an STM32-based pan-tilt gimbal controller for immediate physical camera motion.
- **Manual Positioning:** In **idle** mode, the operator can manually pan and tilt the camera in real time to align the view with the launch rail or expected ascent corridor.
- **Remote Operation:** Core control actions are available through a browser over a local network, allowing safe stand-off operation without internet access.
- **System-State Awareness:** Clear visual feedback on the current state (**idle**, **armed**, **tracking**) to quickly verify readiness during time-critical procedures.
- **Safe Override:** Operator can immediately stop automated motion and return to a safe **idle** state for fault recovery or unexpected target behavior.



Two-Axis Gimbal Prototype

- **Autonomous Vision Tracking:** Once **armed**, RoCam continuously processes the live feed to detect the rocket as a fast-moving target under realistic conditions.
- **Automatic State Transition:** Automatically transitions from **armed** to **tracking** upon valid detection, reducing reaction delay during critical flight phases.
- **Closed-Loop Target Following:** Vision pipeline detections are converted into continuous pan and tilt corrections to keep the rocket centered throughout the flight.
- **Robustness in Challenging Conditions:** Effective against common launch disturbances such as smoke plumes, glare, motion blur, and rapid angular movement.
- **Short-Term Re-Acquisition:** Attempts to re-acquire the target if briefly lost, improving tracking continuity and preserving flight footage without manual intervention.

Computer Vision & Machine Learning



YOLO26s-P2 Training Performance

- **Architecture Upgrade (YOLO26s-P2):** Added a **stride-4, detection head** to quadruple the spatial resolution of the finest feature map, ensuring sensitivity to micro-targets as small as 10×10 pixels.
- **Data Engineering & Hard Negative Mining:** Engineered an automated pipeline to inject **4,000 pure background COCO images** (~25% of the dataset) with empty labels, forcing a harder decision boundary to suppress false positives.
- **Custom Degradation Pipeline:** Integrated a DDP-compatible Albumentations pipeline to simulate real-world flight conditions, including motion blur, sensor noise, dynamic range shifts, and compression artifacts.
- **Curriculum Augmentation Strategy:** Implemented a progressive reduction strategy across training stages—shifting from heavy augmentation (exploration) to minimal perturbation (exploitation).
- **Failure Analysis & Debugging:** Diagnosed a critical "Mosaic Starvation" failure where early stopping preempted the `close_mosaic` phase, trapping the model on $\frac{1}{4}$ -scale downsampled images and crashing mAP_{50-95} to 0.614.
- **Close-Mosaic Fine-Tuning:** Engineered a dedicated recovery stage (`mosaic=0.0`) to expose the model to full-resolution 960×544 frames, rescuing the pipeline and triggering a massive +17.3% metric jump.
- **HPC Distributed Training:** Orchestrated the complete four-stage training pipeline on a **4×NVIDIA H100 PCIe (81 GB)** cluster using Distributed Data Parallel (DDP) and mixed precision.

Final Deployment Metrics: **$mAP_{50} = 0.958$; $mAP_{50-95} = 0.750$**

Our Team

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